Work Item Document

This document presents the current state of all Work Items in the project as of **Monday 2025/10/20 14:19**.

Each Work Items includes details such as their title, current state, and the time of the last edit. For each Work Item, a link to it in the backlog has also been provided to observe the full details. The Work Item hierarchy goes in the order of Project > Epics > Features > User Stories, with all child Work Items under their respective parents.

For the full backlog, please see the following link: 'The Racetrack Game' backlog.

Additional Notes:
As per supervisor recommendations, the wording of user stories has been changed. The current structure will be reviewed in a supervisor meeting and may be adjusted accordingly depending on feedback.

ID	Type	Title	State	Story Points	Created	Last Changed
345	Project	The Racetrack Game	New	0	2025-07-18 10:42:36	2025-09-29 09:10:02
331	Epic	Version 0 (MVP for Core Gameplay)	New	0	2025-07-18 09:30:52	2025-10-20 08:52:18
351	Feature	[V0] - Core	New	0	2025-09-29 12:14:54	2025-10-20 09:06:28
458	User Story	[V0] - The player MUST be able to run and view the game	Requirements	0	2025-10-20 07:46:01	2025-10-20 13:10:25
459	Task	Create "Main Menu" section	New	0	2025-10-20 07:47:17	2025-10-20 07:49:47
461	Task	Show "Main Menu" section on app start	New	0	2025-10-20 07:50:11	2025-10-20 07:50:11

361	Feature	[V0] - Track Viewing & Selection	New	0	2025-10-02 08:51:46	2025-10-20 09:06:33
431	User Story	[V0] - The player MUST be able to view all available tracks	Requirements	0	2025-10-20 06:06:07	2025-10-20 13:10:25
462	Task	Create "Play" option in "Main Menu" section	New	0	2025-10-20 07:52:07	2025-10-20 07:52:47
463	Task	Implement access to "Tracks List" section	New	0	2025-10-20 07:52:26	2025-10-20 07:52:47
464	Task	Create "Tracks List" section UI	New	0	2025-10-20 07:52:44	2025-10-20 07:52:47
465	Task	Implement "Tracks List" section closure	New	0	2025-10-20 07:53:46	2025-10-20 07:53:46
472	Task	Ensure "the test track" is visible in track list as an entry	New	0	2025-10-20 08:06:33	2025-10-20 08:08:01
432	User Story	[V0] - The player MUST be able to choose a track to play	Requirements	0	2025-10-20 06:06:38	2025-10-20 13:10:25
473	Task	Ensure "the test track" is selectable in track list	New	0	2025-10-20 08:06:59	2025-10-20 08:06:59
475	Task	Ensure a race can be started by selecting "the test track" (and generally)	New	0	2025-10-20 08:08:22	2025-10-20 08:11:12
436	User Story	[V0] - The player SHOULD be able to view an existing track's details	Requirements	0	2025-10-20 06:17:16	2025-10-20 13:10:25
467	Task	Implement details displayed on track entry selection	New	0	2025-10-20 07:56:53	2025-10-20 07:56:53

468	Task	Create track details UI	New	0	2025-10-20 07:57:05	2025-10-20 07:57:09
474	Task	Ensure details of "the test track" are visible (and generally)	New	0	2025-10-20 08:07:51	2025-10-20 08:11:26
365	Feature	[V0] - Track Building	New	0	2025-10-02 09:04:00	2025-10-20 09:06:38
434	User Story	[V0] - The player MUST be able to build their own track	Requirements	0	2025-10-20 06:07:49	2025-10-20 13:10:25
469	Task	Implement track representation with splines	New	0	2025-10-20 08:04:10	2025-10-20 08:16:24
470	Task	Implement mask extent and dimension definitions	New	0	2025-10-20 08:05:11	2025-10-20 08:05:11
471	Task	Auto-generate circular track as a default track ("the test track")	New	0	2025-10-20 08:06:04	2025-10-20 08:06:04
480	Task	Create "Track Builder" section UI	New	0	2025-10-20 08:19:42	2025-10-20 08:19:42
481	Task	Implement access to "Track Builder" section	New	0	2025-10-20 08:20:05	2025-10-20 08:20:26
482	Task	Implement "Track Builder" section closure	New	0	2025-10-20 08:20:43	2025-10-20 08:20:43
372	Feature	[V0] - App Settings	New	0	2025-10-02 17:14:21	2025-10-20 12:25:36
435	User Story	[V0] - The player MUST be able to view all their current application settings	Requirements	0	2025-10-20 06:16:12	2025-10-20 13:10:25

455	Task	Create "Settings" section UI	New	0	2025-10-20 07:37:35	2025-10-20 07:49:22
457	Task	Implement "Settings" section closure	New	0	2025-10-20 07:38:20	2025-10-20 07:49:08
460	Task	Implement access to "Settings" section	New	0	2025-10-20 07:48:42	2025-10-20 07:48:57
377	Feature	[V0] - Racing Gameplay	New	0	2025-10-03 06:07:00	2025-10-20 09:06:41
433	User Story	[V0] - The player MUST start a race from the starting line	Requirements	0	2025-10-20 06:07:12	2025-10-20 13:10:25
479	Task	Ensure player race car starts on finish line	New	0	2025-10-20 08:15:14	2025-10-20 08:15:14
447	User Story	[V0] - The player MUST be able to control their car when in a race	Requirements	0	2025-10-20 07:05:54	2025-10-20 13:10:25
489	Task	Ensure track grid is visible over track	New	0	2025-10-20 08:37:35	2025-10-20 13:14:00
490	Task	Ensure only tiles within velocity & acceleration bounds are selectable	New	0	2025-10-20 08:38:15	2025-10-20 08:38:15
555	Task	Implement velocity change with grid tile selection	New	0	2025-10-20 13:13:51	2025-10-20 13:13:57
556	Task	Implement race car transition between tiles	New	0	2025-10-20 13:14:13	2025-10-20 13:14:13
448	User Story	[V0] - The player MUST be able to complete an ongoing race	Requirements	0	2025-10-20 07:06:54	2025-10-20 13:10:25

491	Task	Implement race completion check when crossing finish line	New	0	2025-10-20 08:40:08	2025-10-20 08:40:08
494	Task	End race when player crosses finish line	New	0	2025-10-20 08:43:59	2025-10-20 08:43:59
496	Task	Display race results with completion details	New	0	2025-10-20 08:44:33	2025-10-20 08:44:33
476	User Story	[V0] - The player MUST be able to see their race car on the track	Requirements	0	2025-10-20 08:14:24	2025-10-20 13:10:25
477	Task	Implement race car representation	New	0	2025-10-20 08:14:40	2025-10-20 08:14:44
478	Task	Ensure player race car is visible on track	New	0	2025-10-20 08:14:55	2025-10-20 08:14:55
519	Task	Create sprite for default race car	New	0	2025-10-20 09:32:18	2025-10-20 09:32:23
483	User Story	[V0] - The player MUST be able to see the track they've selected to play	Requirements	0	2025-10-20 08:26:45	2025-10-20 13:10:25
484	Task	Create "Gameplay" section UI	New	0	2025-10-20 08:27:05	2025-10-20 08:27:05
485	Task	Implement "Gameplay" screen closure	New	0	2025-10-20 08:27:46	2025-10-20 08:27:46
486	Task	Ensure track associated with selected track is shown	New	0	2025-10-20 08:28:26	2025-10-20 08:28:26
487	User Story	[V0] - The player MUST be bound to the current track's confines when racing	Requirements	0	2025-10-20 08:29:06	2025-10-20 13:10:25

492	Task	Implement checks to crash player when exceeding track bounds	New	0	2025-10-20 08:42:06	2025-10-20 08:42:06
493	Task	End race when player crashes	New	0	2025-10-20 08:42:24	2025-10-20 08:42:24
495	Task	Display race results popup with crash details	New	0	2025-10-20 08:44:18	2025-10-20 08:44:39
497	User Story	[V0] - The player SHOULD be able to see the results of a race when completing it	Requirements	0	2025-10-20 08:46:07	2025-10-20 13:10:25
498	Task	Create race results popup UI	New	0	2025-10-20 08:46:24	2025-10-20 08:46:36
499	Task	Implement race results closure	New	0	2025-10-20 08:46:48	2025-10-20 08:46:48
332	Epic	Version 2 (AI, Best Line Prediction, and Local Multiplayer)	New	0	2025-07-18 09:32:30	2025-10-20 08:51:50
346	Feature	AI-Controlled Race Car	New	0	2025-07-18 10:50:01	2025-10-02 08:51:53
347	Feature	Best Line Prediction	New	0	2025-07-18 10:50:27	2025-10-02 08:51:53
375	Feature	"Track Options" screen	New	0	2025-10-02 18:06:21	2025-10-12 12:21:57
341	Epic	Version 1 (Building, Extended Gameplay, and Basic Saving)	New	0	2025-07-18 09:40:50	2025-10-20 08:51:41
344	Feature	[V1] - Track Viewing & Selection	New	0	2025-07-18 09:42:55	2025-10-20 09:08:59

405	User Story	[V1] - The player COULD see a preview of the track when selecting it	Requirements	0	2025-10-14 12:34:41	2025-10-20 09:11:58
445	User Story	[V1] - The player MUST be able to view all their saved built tracks	Requirements	0	2025-10-20 06:35:08	2025-10-20 13:10:25
450	User Story	[V1] - The player COULD be able to filter all available tracks by keywords	Requirements	0	2025-10-20 07:29:52	2025-10-20 13:10:25
512	User Story	[V1] - The player COULD be able to filter all available tracks by theirs	Requirements	0	2025-10-20 09:10:14	2025-10-20 13:10:25
364	Feature	[V1] Track Building - Controls	New	0	2025-10-02 09:01:57	2025-10-20 09:36:41
513	User Story	[V1] - The player MUST be able to build their own track	Requirements	0	2025-10-20 09:15:45	2025-10-20 13:10:25
514	Task	Ensure all build controls are accessible in track builder	New	0	2025-10-20 09:18:21	2025-10-20 09:18:21
386	Feature	[V1] - Race Car Selection	New	0	2025-10-12 10:44:08	2025-10-20 12:26:15
516	User Story	[V1] - The player SHOULD be able to view all available race cars	Requirements	0	2025-10-20 09:28:46	2025-10-20 13:10:25
517	User Story	[V1] - The player SHOULD be able to choose a race car to race with	Requirements	0	2025-10-20 09:29:18	2025-10-20 13:10:25
518	User Story	[V1] - The player SHOULD start a race with their selected car	Requirements	0	2025-10-20 09:31:34	2025-10-20 13:10:25
411	Feature	[V1] - Track Management	New	0	2025-10-14 12:48:01	2025-10-20 09:13:37

419	User Story	[V1] - The player SHOULD be able to save copies of tracks they've built	Requirements	0	2025-10-14 13:09:05	2025-10-20 09:29:47
422	User Story	[V1] - The player COULD be able to set tags or identifiers for tracks they've built	Requirements	0	2025-10-14 13:12:29	2025-10-20 12:28:12
439	User Story	[V1] - The player MUST be able to save the track they're building	Requirements	0	2025-10-20 06:27:00	2025-10-20 13:10:25
508	Task	Create save track popup UI	New	0	2025-10-20 09:02:47	2025-10-20 09:03:47
509	Task	Implement providing track details	New	0	2025-10-20 09:03:28	2025-10-20 09:03:37
510	Task	Implement saving track as JSON file to player machine	New	0	2025-10-20 09:04:28	2025-10-20 09:04:28
440	User Story	[V1] - The player SHOULD be able to edit the details of a track they've built	Requirements	0	2025-10-20 06:27:31	2025-10-20 13:10:25
441	User Story	[V1] - The player SHOULD be able to delete a track they've built	Requirements	0	2025-10-20 06:28:24	2025-10-20 13:10:25
446	User Story	[V1] - The player SHOULD be able to set the details of the track they're building	Requirements	0	2025-10-20 06:36:44	2025-10-20 13:10:25
425	Feature	[V1] - Live Race Details	New	0	2025-10-14 13:44:04	2025-10-20 12:26:08
426	User Story	[V1] - The player COULD be able to see their current speed	Requirements	0	2025-10-14 13:44:32	2025-10-20 13:10:25

550	Task	Create speed display UI	New	0	2025-10-20 13:11:24	2025-10-20 13:11:24
551	Task	Implement update speed display with race car velocity	New	0	2025-10-20 13:11:45	2025-10-20 13:11:45
427	User Story	[V1] - The player COULD be able to see the checkpoints they've passed	Requirements	0	2025-10-14 13:44:46	2025-10-20 13:10:25
453	User Story	[V1] - The player COULD be able to view the number of turns taken in a race so far	Requirements	0	2025-10-20 07:34:51	2025-10-20 13:10:25
552	Task	Create total turns UI	New	0	2025-10-20 13:12:09	2025-10-20 13:12:09
553	Task	Implement update turn display on turn taken	New	0	2025-10-20 13:12:25	2025-10-20 13:12:25
537	User Story	[V1] - The player COULD see the path they've taken around the track throughout the race	Requirements	0	2025-10-20 12:21:29	2025-10-20 13:10:25
554	Task	Implement path tracking over turns	New	0	2025-10-20 13:12:46	2025-10-20 13:12:46
522	Feature	[V1] - Track Building - Form	New	0	2025-10-20 09:36:54	2025-10-20 12:25:45
444	User Story	[V1] - The player MUST be able to adjust the track path of a track they're building	Requirements	0	2025-10-20 06:34:19	2025-10-20 13:10:25
500	Task	(Splines) Implement track point creation at location	New	0	2025-10-20 08:56:48	2025-10-20 09:01:19

501	Task	(Splines) Implement moving placed track points	New	0	2025-10-20 08:57:07	2025-10-20 09:01:19
502	Task	(Splines) Implement placed track point deletion	New	0	2025-10-20 08:57:32	2025-10-20 09:01:19
503	Task	(Splines) Render current track point configuration as track	New	0	2025-10-20 08:58:23	2025-10-20 12:29:51
504	Task	(Splines) Display all track points in single list as entries	New	0	2025-10-20 08:58:55	2025-10-20 12:31:24
505	Task	(Splines) Implement track point visibility toggling	New	0	2025-10-20 08:59:39	2025-10-20 12:31:33
506	Task	(Splines) Implement batch track point visibility toggling	New	0	2025-10-20 09:00:17	2025-10-20 12:31:40
507	Task	(Splines) Implement batch deletion of track points	New	0	2025-10-20 09:00:41	2025-10-20 12:31:37
539	Task	(Splines) Implement track point name editing	New	0	2025-10-20 12:31:02	2025-10-20 12:31:27
538	User Story	[V1] - The player COULD be able to define checkpoints for the track	Requirements	0	2025-10-20 12:22:23	2025-10-20 13:10:25
540	Task	(Splines) Implement track point set to checkpoint	New	0	2025-10-20 12:32:44	2025-10-20 12:34:36
541	Task	(Splines) Implement checkpoint indicator for track point	New	0	2025-10-20 12:33:36	2025-10-20 12:34:44
542	Task	(Splines) Implement checkpoint deletion	New	0	2025-10-20 12:34:11	2025-10-20 12:34:52

523	Feature	[V1] - Track Building - Materials	New	0	2025-10-20 09:37:24	2025-10-20 09:38:25
454	User Story	[V1] - The player SHOULD be able to adjust a placed material on a track they're building	Requirements	0	2025-10-20 07:36:08	2025-10-20 13:10:25
544	Task	Implement track material selection on track	New	0	2025-10-20 12:35:49	2025-10-20 12:35:49
548	Task	Implement track material selection from placed materials list	New	0	2025-10-20 12:39:14	2025-10-20 12:39:14
549	Task	Implement movement for selected track material	New	0	2025-10-20 12:39:36	2025-10-20 12:39:36
511	User Story	[V1] - The player SHOULD be able to place a material on the track they're building	Requirements	0	2025-10-20 09:05:23	2025-10-20 13:10:25
531	Task	Implement track material entry selection	New	0	2025-10-20 12:12:04	2025-10-20 12:35:36
543	Task	Implement material placement at location	New	0	2025-10-20 12:35:25	2025-10-20 12:35:25
515	User Story	[V1] - The player SHOULD be able to remove a placed material from a track they're building	Requirements	0	2025-10-20 09:24:38	2025-10-20 13:10:25
521	User Story	[V1] - The player SHOULD be able to see all available materials to place on the track they're building	Requirements	0	2025-10-20 09:34:12	2025-10-20 13:10:25

530	Task	Create track materials list UI	New	0	2025-10-20 12:11:12	2025-10-20 12:15:40
532	Task	Implement track materials list visibility toggling	New	0	2025-10-20 12:12:25	2025-10-20 12:15:40
533	Task	Implement track material types	New	0	2025-10-20 12:13:52	2025-10-20 12:15:40
545	User Story	[V1] - The player SHOULD be able to see a list of all materials placed on the track	Requirements	0	2025-10-20 12:36:20	2025-10-20 13:10:25
546	Task	Create placed track materials list UI	New	0	2025-10-20 12:38:14	2025-10-20 12:38:14
547	Task	Implement placed track materials visibility toggling	New	0	2025-10-20 12:38:49	2025-10-20 12:38:49
524	Feature	[V1] - Track Building - Decorations	New	0	2025-10-20 09:38:15	2025-10-20 12:25:50
437	User Story	[V1] - The player COULD be able to place decorations on a track they're building	Requirements	0	2025-10-20 06:20:46	2025-10-20 13:10:25
451	User Story	[V1] - The player COULD be able to remove a placed decoration from a track they're building	Requirements	0	2025-10-20 07:31:13	2025-10-20 13:10:25
452	User Story	[V1] - The player COULD be able to adjust a placed decoration on a track they're building	Requirements	0	2025-10-20 07:32:19	2025-10-20 13:10:25
520	User Story	[V1] - The player COULD be able to see all available decorations to place on the track they're building	Requirements	0	2025-10-20 09:33:45	2025-10-20 13:10:25

525	Feature	[V1] - Track Building - Obstacles	New	0	2025-10-20 09:39:08	2025-10-20 09:39:20
526	User Story	[V1] - The player SHOULD be able to see all available obstacles to place on the track they're building	Requirements	0	2025-10-20 12:06:30	2025-10-20 13:10:25
557	Task	Create placed obstacles list UI	New	0	2025-10-20 13:18:04	2025-10-20 13:18:04
558	Task	Implement obstacle additions to list on placement	New	0	2025-10-20 13:18:22	2025-10-20 13:18:22
527	User Story	[V1] - The player SHOULD be able to place a material on the track they're building	Requirements	0	2025-10-20 12:08:37	2025-10-20 13:10:25
528	User Story	[V1] - The player SHOULD be able to adjust a placed obstacle on a track they're building	Requirements	0	2025-10-20 12:09:48	2025-10-20 13:10:25
529	User Story	[V1] - The player SHOULD be able to remove a placed obstacle from a track they're building	Requirements	0	2025-10-20 12:10:09	2025-10-20 13:10:25
535	Feature	[V1] - Racing Gameplay	New	0	2025-10-20 12:17:40	2025-10-20 12:17:53
534	User Story	[V1] - The player SHOULD be affected by track materials when driving over them	Requirements	0	2025-10-20 12:16:25	2025-10-20 13:10:25
536	User Story	[V1] - The player SHOULD be considered crashed when driving into track obstacles	Requirements	0	2025-10-20 12:18:49	2025-10-20 13:10:25

366	Epic	Version 3 (Multiplayer, Customisation, and Extended Data Management)	New	0	2025-10-02 09:13:07	2025-10-20 08:51:55
395	Feature	Tracks Database	New	0	2025-10-14 12:26:21	2025-10-14 12:28:00
396	Feature	Player Multiplayer	New	0	2025-10-14 12:26:56	2025-10-14 12:28:00
397	Feature	Car Customisation	New	0	2025-10-14 12:27:22	2025-10-14 12:28:09
398	Feature	Player Data Database	New	0	2025-10-14 12:27:54	2025-10-14 12:28:00